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Analysis of LTI Systems in the Frequency Domain

An LTI System is characterized by its impulse response $h(t)$ in time domain or equivalent by its frequency response $H(f)$ in the frequency domain.

$$x(t) \quad \rightarrow \quad y(t) = h(t) * x(t)$$

$$x(t) \overset{\mathcal{F}}{\circ\!-\!\circ} X(f) \qquad y(t) \overset{\mathcal{F}}{\circ\!-\!\circ} Y(f) = H(f) \cdot X(f)$$

Magnitude Response, Phase Response

Frequency response: $H(f) = |H(f)| e^{j\varphi(f)} = |H(f)| e^{-jb(f)}$

Magnitude response: $|H(f)| = \sqrt{\{\mathrm{Re}\{H(f)\}\}^2 + \{\mathrm{Im}\{H(f)\}\}^2}$

Phase response: $\varphi(f) = \sphericalangle\{H(f)\} = \arg\{H(f)\} = \arctan\left(\frac{\mathrm{Im}\{H(f)\}}{\mathrm{Re}\{H(f)\}}\right) = -b(f)$

Magnitude response → even $|H(f)| = |H(-f)|$

Phase response → odd $\varphi(f) = -\varphi(-f)$

$$|Y(f)| = |H(f)|\,|X(f)|$$

$$\varphi_Y(f) = \sphericalangle\{H(f)\} + \sphericalangle\{X(f)\}$$

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LTI-Systeme: Transmission Factor, Attenuation, Gain, Phase

$$H(f) = |H(f)| e^{j\varphi(f)} = e^{-g(f)} = e^{-a(f)} e^{-jb(f)}$$

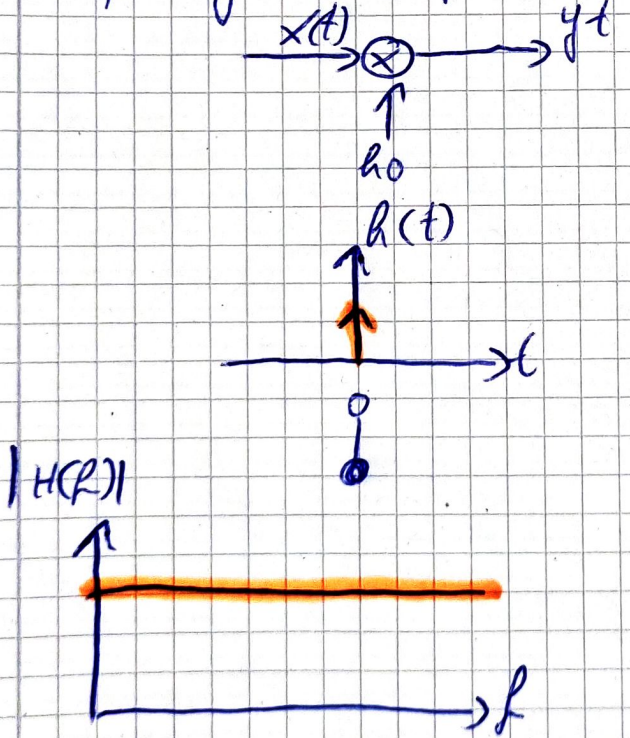
Transmission factor: $g(f) = -\log H(f) = \underbrace{-\log|H(f)|}_{a(f)} \underbrace{-j\varphi(f)}_{-b(f)}$

Attenuation in dB: $a(f) = -20\log_{10}|H(f)|\,dB = -10\log_{10}|H(f)|^2\,dB$

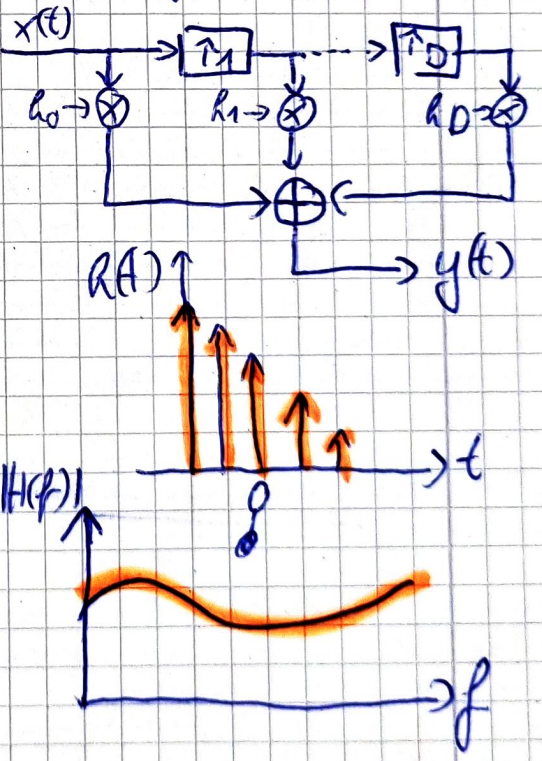
Gain in dB: $-a(f)$

Frequency-Flat & Frequency Selective LTI Systems

Frequency Flat system:



Frequency selective system:



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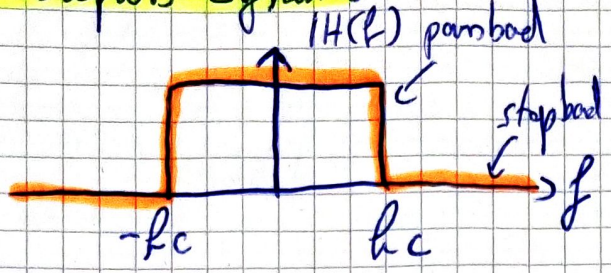
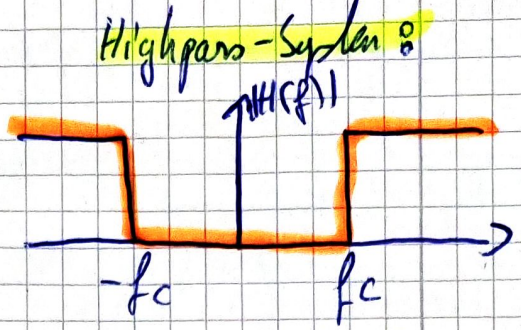
Basic Types of Systems (Filters)

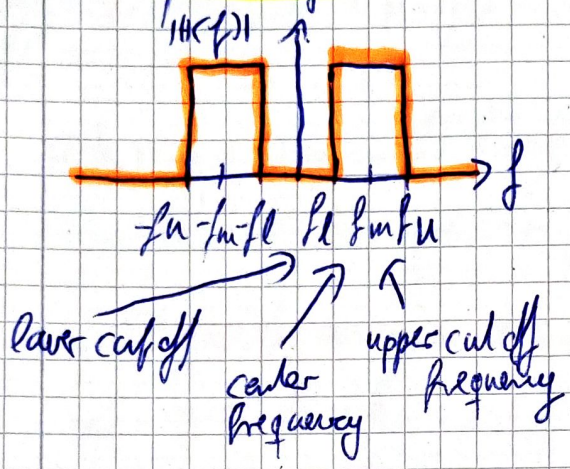
Lowpass-System:



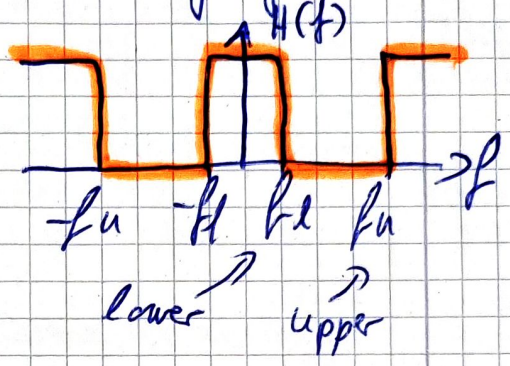
Highpass-System:



Bandpass-System:



Bandstop System:



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LTI System: Phase Delay

$$X(f) = \delta(f - f_0) \quad \circ\!\!-\!\!\bullet \quad x(t) = e^{j 2\pi f_0 t}$$

$$Y(f) = |H(f_0)| \, \delta(f - f_0) \, e^{-j 2\pi f_0 t_p(f_0)} \quad \circ\!\!-\!\!\bullet \quad y(t) = |H(f_0)| \, e^{j 2\pi f_0 (t - t_p(f_0))}$$

- **Phase delay:** $t_p(f) = \frac{-\varphi(f)}{2\pi f} = \frac{b(f)}{2\pi f}$

The phase delay is the delay caused by the system to an oscillation at frequency f.

↳ Problem many systems has the same phase delay

← more suitable

- **Group delay:** $t_g(f) = \frac{-1}{2\pi} \frac{d}{df} \varphi(f) = \frac{1}{2\pi} \frac{d}{df} b(f)$

The group delay $t_g(f)$ is the delay caused by the system to the envelope of a narrowband signal.
The delay caused to an oscillation at the carrier frequency f_0 is given by the phase delay.

Linear phase Systems

Linear phase systems have a constant group delay, i.e. all frequency components of the input signal are delayed by the same amount

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Group delay ≙ Verschiebung der Umhüllenden $a(t)$
Phase delay ≙ Verschiebung des Signales $x(t)$

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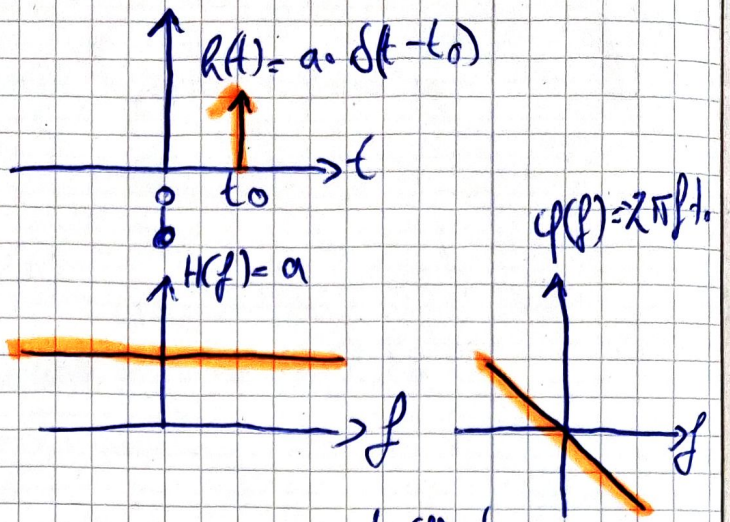
Distortion-Free Systems

Impulse response

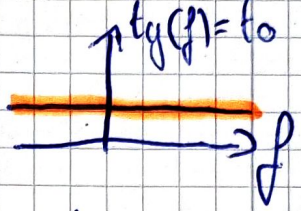
$h(t) = a \cdot \delta(t - t_0)$

Frequency response

$H(f) = a e^{-j 2\pi f t_0}$



Group delay: $t_g(f) = \frac{-1}{2\pi} \frac{d}{df} \cdot \varphi(f) = t_0$



A distortion-free system caused only a delay and an amplitude scaling to the input signal. The magnitude response is constant, the phase is linear and the group delay is constant

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Group & Phase Delay

An analog filter with the following frequency response is given:

$$H(f) = \frac{1 - j2\pi f T}{1 + j2\pi f T}$$

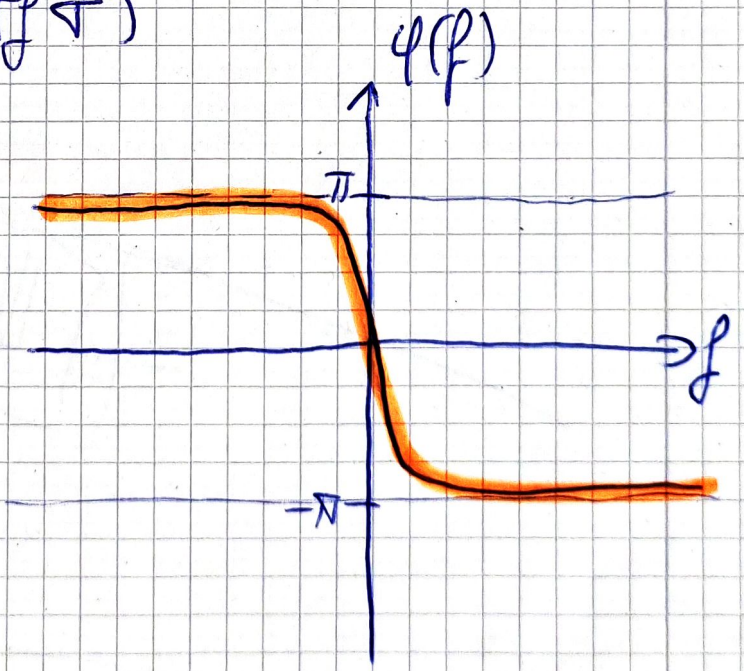
a) Determine & draw the magnitude respond $|H(f)|$ & phase respond $\varphi(f)$

$$|H(f)| = \frac{|H_z(f)|}{|H_n(f)|} = \frac{\sqrt{1^2 + 2\pi^2 f^2 T^2}}{\sqrt{1^2 + 2\pi^2 f^2 T^2}} = 1$$

$$\varphi(f) = \varphi_z(f) - \varphi_n(f) = \arctan\left(\frac{-2\pi f T}{1}\right) - \arctan\left(\frac{2\pi f T}{1}\right)$$

$$= \arctan(-2\pi f T) - \arctan(2\pi f T)$$

$$= -2\arctan(2\pi f T)$$



b) Allpass-Filter

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Group & Phase Delay

c) Calculate the phase delay $t_p(f)$ and the group delay $t_g(f)$

$$t_p(f) = \frac{-\varphi(f)}{2\pi f} = \frac{2\arctan(2\pi f T)}{2\pi f} = \frac{\arctan(2\pi f T)}{\pi f}$$

$$t_g(f) = \frac{-1}{2\pi}\frac{d}{df}\varphi(f) = \frac{\frac{2 \cdot 2\pi T}{1+4\pi^2 f^2 T^2}}{2\pi} = \frac{2T}{1+4\pi^2 f^2 T^2}$$

$$\frac{d}{dx}\arctan(x) = \frac{1}{1+x^2}$$

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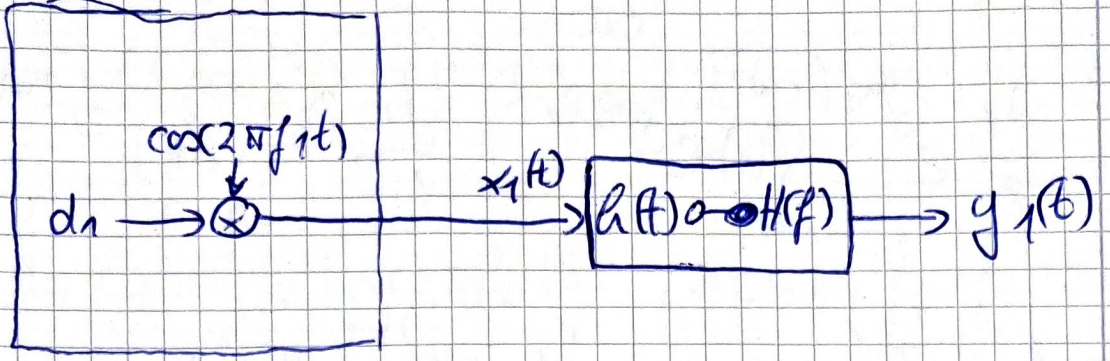
Multicarrier Transmission Part I

$$x_1(t) = d_1 \cos(2\pi f_1 t) \quad d_1 \in \mathbb{R}, f_1 > 0$$



the frequency response $H(f)$ is real & even in f

a) According to the symetry properties of the Fourier transform, an even frequency respose $H(f)$ leads to an even impulse respose $h(t)$, i.e $h(t) = h(-t)$.
Thus the impulse respose has non-zero values for $t < 0$ and → non causal

b) Show $y_1(t) = H(f_1) \cdot x_1(t)$

$$y_1(t) = h(t) * x_1(t) = \int_{-\infty}^{\infty} h(\tau)\, x_1(t-\tau)\, d\tau$$

$$= \int_{-\infty}^{\infty} h(\tau) \cdot \frac{d_1}{2}\left(e^{j2\pi f_1(t-\tau)} + e^{-j2\pi f_1(t-\tau)}\right) d\tau$$

$$\frac{d_1}{2} e^{j2\pi f_1 t} \left(\int_{-\infty}^{\infty} h(\tau) \left(e^{-j2\pi f_1 \tau} + e^{j2\pi f_1 \tau} \right) d\tau \right)$$

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Multicarrier Transmission Pt. I

b) $y_1(t) = \frac{d_1}{2}\left(e^{j2\pi f_1 t}\int_{-\infty}^{\infty} h(\tau)e^{-j2\pi f_1 \tau}d\tau + e^{-j2\pi f_1 t}\int_{-\infty}^{\infty} h(\tau)e^{-j2\pi(-f_1)\tau}d\tau\right)$

$= \frac{d_1}{2}\left(e^{j2\pi f_1 t} + e^{-j2\pi f_1 t}\right) H(f_1)$

$x_2(t) = d_2 \cos(2\pi f_2 t)$, $d_2 \in \mathbb{R}$, $f_2 > f_1 > 0$

$x(t) = x_1(t) + x_2(t)$

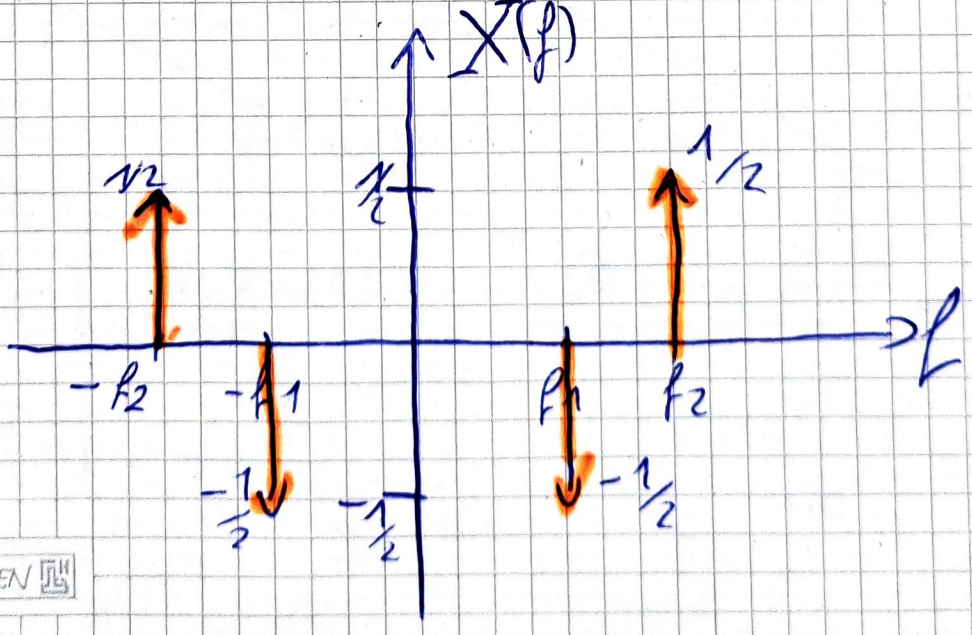
c) Calculate $X(f)$

$x(t) = d_1 \cos(2\pi f_1 t) + d_2 \cos(2\pi f_2 t)$

$\circ\!-\!\bullet$

$X(f) = \frac{d_1}{2}[\delta(f+f_1) + \delta(f-f_1)] + \frac{d_2}{2}[\delta(f+f_2) + \delta(f-f_2)]$

d) $d_1 = -1$, $d_2 = 1$, $f_2 > f_1$



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Multicarrier Transmission Part 1

e) $T = \frac{1}{f_1}$

$$r_{x_1x_2}^{P}(\tau) = f_1 \int_0^{1/f_1} d_1 \cos(2\pi f_1 t)\, d_2 \cos(2\pi f_1 (t+\tau))\, dt$$

$$r_{x_1x_2}^{P}(\tau=0) = d_1 d_2 \frac{f_2^2}{f_2^2 - f_1^2} \operatorname{si}\left(2\pi \frac{f_2}{f_1}\right) \quad !$$

$$= \langle x_1(t), x_2(t) \rangle$$

$f_2 > f_1 > 0$

Goal: $x_1(t)$ & $x_2(t)$ orthogonal within the time period T_0 !

$$\langle x_1(t), x_2(t) \rangle = r_{x_1x_2}^{P}(0) = 0$$

f) When are $x_1(t)$ & $x_2(t)$ orthogonal?

$$\langle x_1(t), x_2(t) \rangle = \int_{-\infty}^{\infty} x_1^*(t)\, x_2(t)\, dt = \int_{-\infty}^{\infty} d_1 d_2 \cos(2\pi f_1 t) \cos(2\pi f_2 t)\, dt$$

$$= \int_{-\infty}^{\infty} \frac{d_1 d_2}{2} \left(\cos(2\pi (f_1 - f_2) t) + \cos(2\pi (f_1 + f_2) t) \right) dt$$

$$= \frac{d_1 d_2}{2} \int_{-\infty}^{\infty} \cos(2\pi t) + \cos(6\pi t)\, dt$$

$$= \frac{d_1 d_2}{2} \left(\left[\frac{\sin(2\pi t)}{2\pi} \right]_{-\infty}^{\infty} + \left[\frac{\sin(2\pi (3t))}{2\pi} \right]_{-\infty}^{\infty} \right) = 0$$

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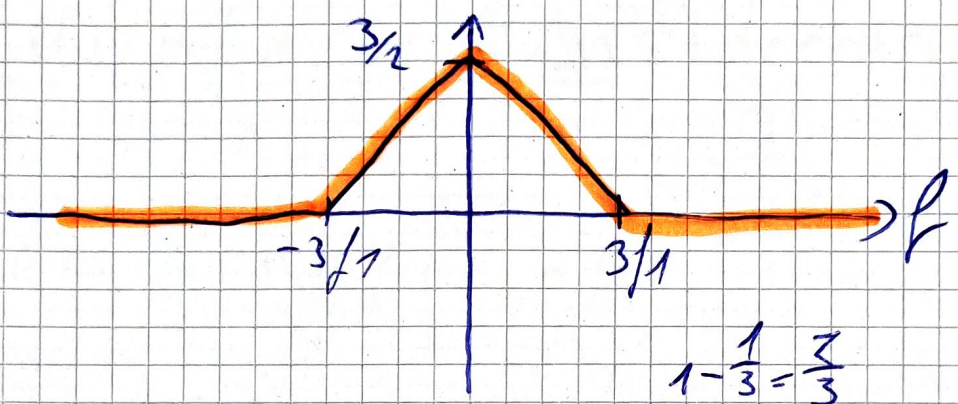
Multicarrier Transmission Part

f) $\langle x_1(t), x_2(t) \rangle = \langle x_1(t), x_2(t+0) \rangle$

$= r_{x_1 x_2}^{P}(0) = d_1 d_2 \frac{4f_1}{4f_1 - f_1} si\left(2\pi \frac{2f_1}{4f_1}\right)$

$= d_1 d_2 \frac{4}{3} si(4\pi) = 0$

g) $H(f) = \frac{3}{2} \Lambda\left(\frac{f}{3f_1}\right)$



$1 - \frac{1}{3} = \frac{2}{3}$

h) $y_1(t) = H(f_1) x_1(t) = \frac{3}{2} \Lambda\left(\frac{1}{3}\right) \cdot d_1 \cos(2\pi f_1 t)$

$1 - \frac{2}{3} = \frac{1}{3}$

$y_2(t) = H(f_2) x_2(t) = \frac{3}{2} \Lambda\left(\frac{2}{3}\right) d_2 \cos(2\pi f_2 t)$

$= \frac{1}{2} d_2 \cos(2\pi f_2 t)$

$y(t) = y_1(t) + y_2(t) = d_1 \cos(2\pi f_1 t) + \frac{1}{2} d_2 \cos(2\pi f_2 t)$

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Multicarrier Transmit Part II

i) What if $f_2' = 3f_1$

$$\langle x_1(t), x_2(t+0) \rangle = r^{P}_{x_1 x_2}(0) = d_1 d_2 \frac{2f_1}{2f_1 - f_1} \operatorname{si}(2\pi \frac{2}{2} \frac{3f_1}{f_1})$$

$$= 0 \rightsquigarrow \text{orthogonal}$$

Oral

$$y_2(t) = H(f_2) x_2(t) = \frac{3}{2} \Lambda(\frac{3}{3}) d_2 \cos(2\pi f_2 t)$$

$$\hookrightarrow 0$$

$$y_1(t) = d_1 \cos(2\pi f_1 t) = y(t)$$

$\hookrightarrow y_2(t)$ is not transmitted

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Laplace Transform

$$X_L(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad ; s = \alpha + j\omega = \alpha + j2\pi f$$

$$x(t) \circ\!\!-\!\!\bullet\!\!\!\bullet\; X(s)$$

The Laplace Transform exists if both Integrals exists

$$\int_{-\infty}^{0} x(t) e^{-st} dt \quad \& \quad \int_{0}^{\infty} x(t) e^{-st} dt$$

$$x(t) = \frac{1}{2\pi j} \int_{a_c - j\infty}^{a_c + j\infty} X_L(s) e^{st} dt \quad \text{// inverse laplace}$$

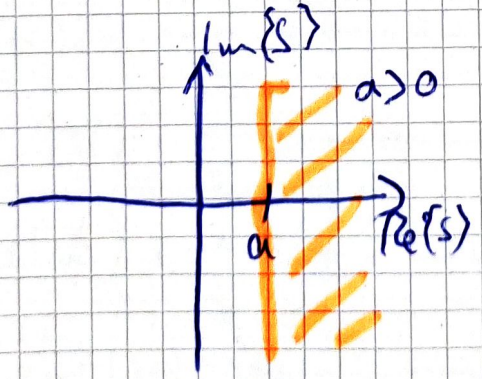
Example:

$$x(t) = e^{a|t|} = \begin{cases} e^{-at}, & t < 0 \\ e^{at}, & t \geq 0 \end{cases} \quad \overset{L}{\circ\!\!-\!\!\bullet\!\!\!\bullet} \quad X_L(s) = \frac{1}{s-a} + \frac{-1}{s+a}$$

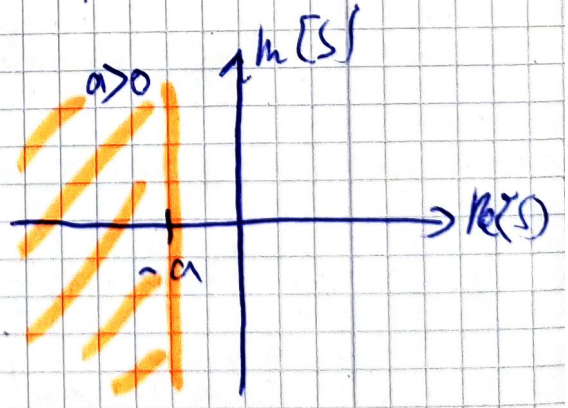
$$X_{L+}(s) = \int_{0}^{\infty} x(t) e^{-st} dt = \frac{1}{s-a} \quad ; \quad X_{L-}(s) = \int_{-\infty}^{0} x(t) e^{-st} dt = \frac{-1}{s+a}$$

Convergence Region

$Re\{s\} > a$



$Re\{s\} < -a$



$\Rightarrow$ no common convergence region $a > 0$

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Laplace Transform

Laplace Transform

$$X_L(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt$$

Fourier Transform

$$X_F(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi f t} dt$$

$$= X_L(s)\big|_{s=j2\pi f}$$

If the convergence region of the Laplace Transform includes the imaginary axis both Laplace transform and Fourier transform exists and are related by

$$X_F(f) = X_L(s)\big|_{s=j2\pi f}$$

Laplace Transform of Discrete Time Signals

The information about the time domain signal $x_L(t)$ is periodically repeated.

Some Rules

$\delta(t) \circ\!\!-\!\!\bullet\; X(s) = 1, \quad s \in \mathbb{C}$

$x(t) = 1 \circ\!\!-\!\!\bullet$ does not exists

$\varepsilon(t) \circ\!\!-\!\!\bullet\; \frac{1}{s}, \quad \mathrm{Re}\{s\} > 0$

$\mathrm{rect}\left(\frac{t}{T}\right) \circ\!\!-\!\!\bullet\; \frac{1}{s}\left(e^{sT/2} - e^{-sT/2}\right), \quad s \in \mathbb{C}$

$\mathrm{sgn}(t) \circ\!\!-\!\!\bullet$ does not exists

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Properties of Laplace Transform

$\sum a_i x_i(t)$	$\circ\!-\!\bullet$	$\sum_i a_i X_i(s)$, $K \supseteq \bigcap_i K_i$
$x(t-t_0)$	$\circ\!-\!\bullet$	$e^{-st_0} X(s)$
$e^{at} x(t)$	$\circ\!-\!\bullet$	$X(s-a)$
$\int_{-\infty}^{t} x(t)\,dt$	$\circ\!-\!\bullet$	$\frac{1}{s} X(s)$
$x(at)$	$\circ\!-\!\bullet$	$\frac{1}{\lvert a\rvert} X\left(\frac{s}{a}\right)$; $K \cdot a$
$x^*(t)$	$\circ\!-\!\bullet$	$X^*(s^*)$; $K = K$
$x(-t)$	$\circ\!-\!\bullet$	$X(-s)$; $-K$
$x^*(-t)$	$\circ\!-\!\bullet$	$X^*(-s)$; $-K$
$x(t) * y(t)$	$\circ\!-\!\bullet$	$X(s)\,Y(s)$, $K \supseteq K_x \cap K_y$
$x(t) \cdot y(t)$	$\circ\!-\!\bullet$	$\frac{1}{2\pi j} \int_{\alpha-j\infty}^{\alpha+j\infty} X(\xi) Y(s-\xi)\,d\xi$, -11-
$r_{xy}^{E}(\tau)$	$\circ\!-\!\bullet$	$X^*(-s^*)\,Y(s)$, $K \supseteq -K_x \cap K_y$
$\frac{d}{dt} x(t)$	$\circ\!-\!\bullet$	$s\,X(s)$
$-t\,x(t)$	$\circ\!-\!\bullet$	$\frac{d}{ds} X(s)$
$x(t) = e^{at} \varepsilon(t)$	$\circ\!-\!\bullet$	$X_L(s) = \frac{1}{s-a}$ → single pole.
$x(t) = t^m e^{at} \varepsilon(t)$	$\circ\!-\!\bullet$	$X_L(s) = \frac{m!}{(s-a)^{m+1}}$ → m+1-any pole
$e^{j2\pi f_0 t} \varepsilon(t)$	$\circ\!-\!\bullet$	$\frac{1}{s - j2\pi f_0}$, $\mathrm{Re}\{s\} > 0$

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Fourier Transform & Laplace Transform

a) $x(t) = 1 \quad \circ\!\!-\!\!\overset{F}{-}\!\!\bullet \quad \delta(f)$

$$X_L(s) = \int_{-\infty}^{\infty} e^{-st}\,dt = \int_{-\infty}^{\infty} e^{-at} e^{-j\omega t}\,dt$$

$$= \frac{1}{-s}\left[e^{-st}\right]_{-\infty}^{\infty} = \frac{1}{-s}\left[e^{-at} e^{-j\omega t}\right]_{-\infty}^{\infty}$$

$$= -\frac{1}{s}\left[e^{-a\infty} e^{-j\omega\infty} - e^{a\infty} e^{j\omega\infty}\right]$$

$a < 0$ ↳ ∞ & ↳ 0 ↯

$a > 0$ ↳ 0 & ↳ ∞

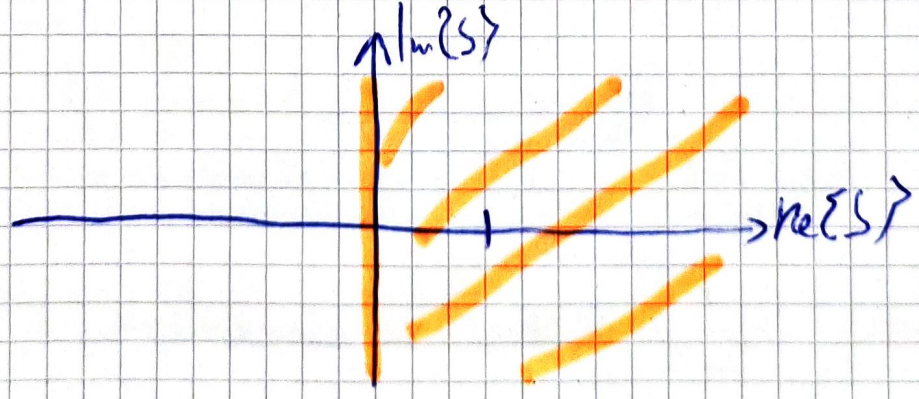
$a = 0$ ↳ undefined

→ Laplace Transform does not exist

b) $\varepsilon(t) \quad \circ\!\!-\!\!\overset{F}{-}\!\!\bullet \quad X_F(f) = \frac{1}{2}\delta(f) + \frac{1}{j2\pi f}$

$$X_L(s) = \int_{-\infty}^{\infty} \varepsilon(t) e^{-st}\,dt = \int_{0}^{\infty} e^{-st}\,dt = \frac{-1}{s}\left[e^{-st}\right]_0^{\infty}$$

$$= \frac{-1}{s}[0 - 1] = \frac{1}{s} \qquad \mathrm{Re}\{s\} > 0$$



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Fourier Transform & Laplace Transform

c) $x(t) = \delta(t) \circ\!\!-\!\!\bullet^{F} 1 = X_F(f)$

$$X_L(s) = \int_{-\infty}^{\infty} \delta(t) e^{-st}\, dt = \int_{-\infty}^{\infty} \delta(t) e^{-s\cdot 0}\, dt = 1, \quad s \in \mathbb{C}$$

($\delta(t) \hookrightarrow 1$)

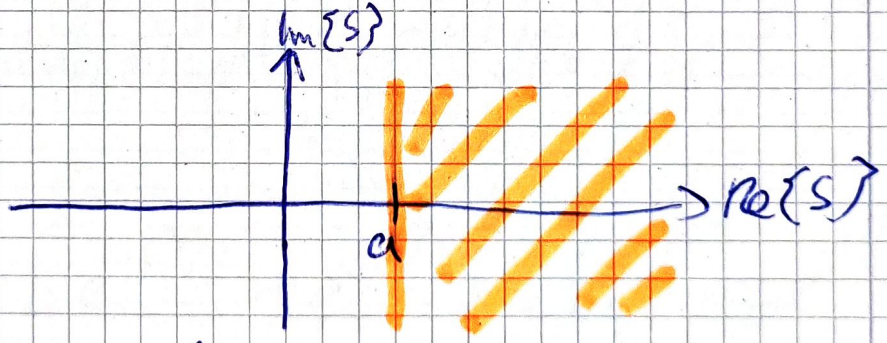
d) $x(t) = e^{at}\varepsilon(t) \circ\!\!-\!\!\bullet^{F}$

$$X_F(f) = \int_{-\infty}^{\infty} \varepsilon(t) e^{at} e^{-j2\pi f t}\, dt = \int_{0}^{\infty} e^{(a-j2\pi f)t}\, dt \Rightarrow \text{does not exist!}$$

no Energy signal

$$X_L(s) = \int_{-\infty}^{\infty} \varepsilon(t) e^{at} e^{-st}\, dt = \int_{0}^{\infty} e^{at} e^{-st}\, dt$$

$$= \frac{1}{a-s}\left[e^{(a-s)t}\right]_0^{\infty} = \frac{1}{a-s}[0-1] = \frac{1}{s-a}, \quad \mathrm{Re}\{s\} > a$$



$$X_F(f) = \int_{-\infty}^{\infty} e^{at} \varepsilon(t) e^{-j2\pi f t}\, dt = \int_{0}^{\infty} e^{(a-j2\pi f)t}\, dt$$

$$= \frac{1}{j2\pi f - a}, \quad a < 0$$

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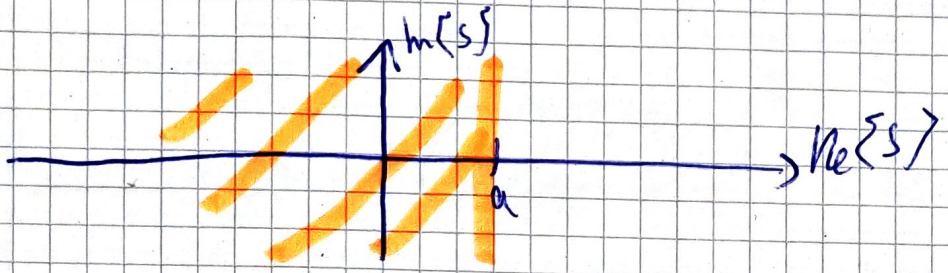
Fourier-Transform & Laplace-Transform

e) $x(t) = -e^{at}\,\varepsilon(-t)$

$X_F(f) = \frac{1}{j2\pi f - a}$, $a \neq 0$

$X_L(s) = \int_{-\infty}^{\infty} -e^{at}\,\varepsilon(-t)\,e^{-st}\,dt = \int_{-\infty}^{0} -e^{at-st}\,dt$

$= \frac{1}{a-s}\left[1-0\right] = \frac{1}{s-a}$, $\mathrm{Re}\{s\} < a$



f) $x(t) = \mathrm{rect}\left(\frac{t}{T}\right) \circ\!\!-\!\!\bullet\; T\,\mathrm{si}(\pi T f) = X_F(f)$

$X_L(s) = \int_{-\infty}^{\infty} \mathrm{rect}\left(\frac{t}{T}\right) e^{-st}\,dt = \int_{-T/2}^{T/2} e^{-st}\,dt = \frac{-1}{s}\left[e^{-st}\right]_{-T/2}^{T/2}$

$= \frac{-1}{s}\left[e^{-sT/2} - e^{sT/2}\right] = \frac{1}{s}\left[e^{sT/2} - e^{-sT/2}\right]$

$s \in \mathbb{C}$

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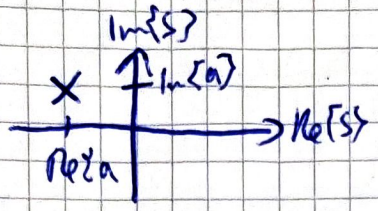
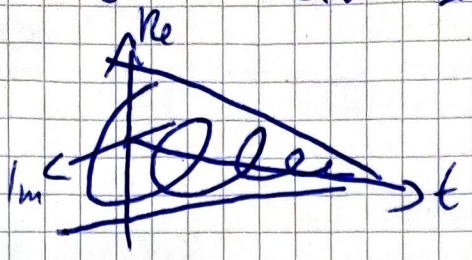
Signale & Systeme

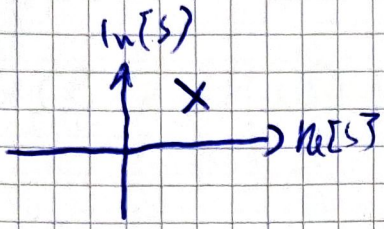
Causal Complex Exponential Function

$$x(t) = e^{at}\,\varepsilon(t) = e^{\mathrm{Re}\{a\}t}\,e^{j\,\mathrm{Im}\{a\}t}\,\varepsilon(t) \circ\!\!-\!\!\bullet\, X(s) = \frac{1}{s-a}$$

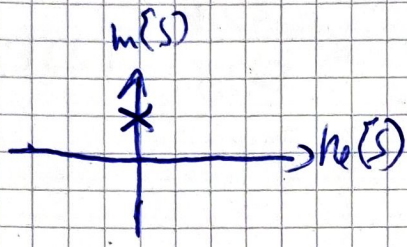




$\mathrm{Re} < 0$



$\mathrm{Re} > 0$



$\mathrm{Re} = 0$

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Signale & Systeme

Analysis of LTI Systems in the s-Domain

$$x(t) \circ\!\!-\!\!\bullet X(s) \quad \Rightarrow \quad y(t) = h(t) * x(t) \circ\!\!-\!\!\bullet Y(s) = H(s) \cdot X(s)$$

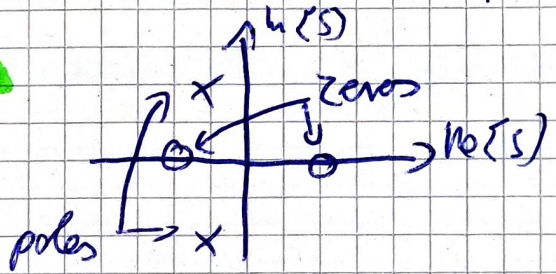
$H(s)$ in the s-domain

$$H(f) = H(s)\big|_{s=j2\pi f}$$

$$H(s) = \frac{F(s)}{B(s)} = \frac{f_M \prod_{k=1}^{M} (s - s_{0,k})}{b_N \prod_{l=1}^{N} (s - s_{\infty,l})}$$

($\rightarrow$ zeros; $\rightarrow$ poles)

Pole-Zero Plot:



Magnitude response:

$$|H(f)| = |H(s)|_{s=j2\pi f}| = \left|\frac{f_M}{b_N}\right| \frac{\prod_{k=1}^{M} |j2\pi f - s_{0,k}|}{\prod_{l=1}^{N} |j2\pi f - s_{0,l}|}$$

Phase response:

$$\varphi(f) = \angle\left(\frac{f_M}{b_N}\right) + \sum_{k=1}^{M} \angle(j2\pi f - s_{0,k}) - \sum_{l=1}^{N} \angle(j2\pi f - s_{0,l})$$

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Signals & Systems

LTI Systems:

An LTI system is characterized by its impulse response $h(t)$ in the time domain or equivalent by its transfer function $H(f)$ in the frequency domain or $H(s)$ in the s-domain

$$H(s) = \frac{Y(s)}{X(s)}$$

Except for a factor μ_1/b_N, a rational transfer function $H(s)$ can be completely described by its pole-zero plot.

Rational Function:

Rational Functions $R(x)$ with real coefficients $\mu_i, b_N \in \mathbb{R}$ have only real or conjugate complex pairs of zeros & poles!

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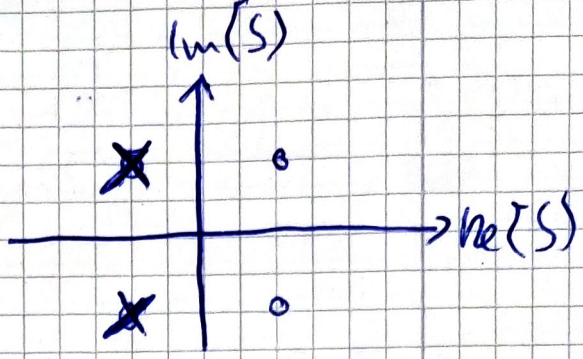
Signale & Systeme

Allpass Filters

$$|H(f)| = \left|\frac{\prod_M}{\prod_N}\right| = \text{const}$$

↳ $s_{0,k} = -s^*_{\infty,k}$



An allpass filter is characterized by a constant magnitude response. It only causes a phase shift and hence a frequency-selective delay to the output.

Poles & Zeros are symmetric: $s_{0,k} = -s^*_{\infty,k}$

Minimum Phase & Maximum Phase Filters

Minimum phase filter → all zeros in the left half plane

Maximum phase filter → all zeros in the right half plane

↳ Both same Magnitude response

⇒ Minimum phase filters have minimum group delay $t_g(f)$

Mixed phase filters: can be decomposed into the cascadation of a minimum phase subsystem and an allpass subsystem!

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Signale & Systeme

Laplace & Inverse Laplace Transform

a) $x(t) = e^{-2t}\,\varepsilon(t) + e^{-3t}\,\varepsilon(t)$

$$e^{at}\,\varepsilon(t) \circ\!\!-\!\!\bullet\; X_L(s) = \frac{1}{s-a}, \quad \mathrm{Re}\{s\} > a$$

$$X_L(s) = \frac{1}{s+2} + \frac{1}{s+3}, \quad \mathrm{Re}\{s\} > -2$$

b) $x(t) = e^{-3t}\,\varepsilon(t) + e^{2t}\,\varepsilon(-t)$

$$-e^{at}\,\varepsilon(-t) \circ\!\!-\!\!\bullet\; X_L(s) = \frac{1}{s-a}, \quad \mathrm{Re}\{s\} < a$$

$$X_L(s) = \frac{1}{s+3} + \frac{1}{s-2}, \quad -3 < \mathrm{Re}\{s\} < 2$$

c) $x(t) = e^{2t}\,\varepsilon(t) + e^{-3t}\,\varepsilon(-t)$

$$X_L(s) = \frac{1}{s-2} + \frac{1}{s+3}, \quad 2 < \mathrm{Re}\{s\}$$
$$\wedge\ \mathrm{Re}\{s\} < -3$$

↯

Wichtig:

$$e^{at}\,\varepsilon(t) \circ\!\!-\!\!\bullet\; \frac{1}{s-a}, \quad \mathrm{Re}\{s\} > \mathrm{Re}\{a\}$$

$$-e^{at}\,\varepsilon(-t) \circ\!\!-\!\!\bullet\; \frac{1}{s-a}, \quad \mathrm{Re}\{s\} < \mathrm{Re}\{a\}$$

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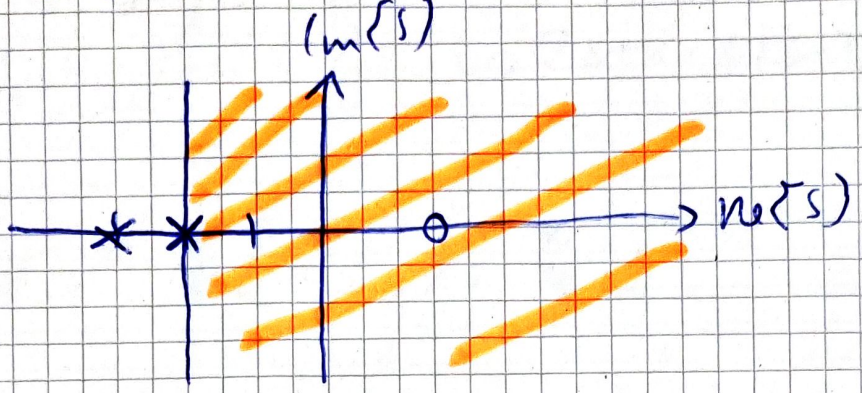
Signale & Systeme

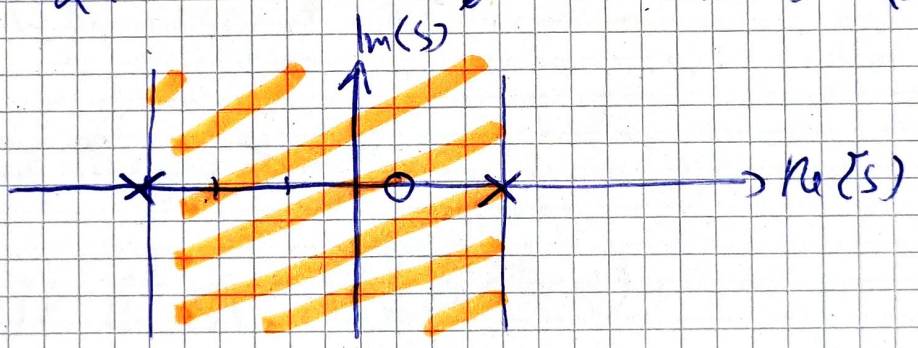
Laplace & Inverse Laplace Transform

a) $X_a(s) = \frac{1}{s+2} + \frac{1}{s+3} = \frac{s+3+s+2}{(s+2)(s+3)} = \frac{2s+5}{(s+2)(s+3)}$

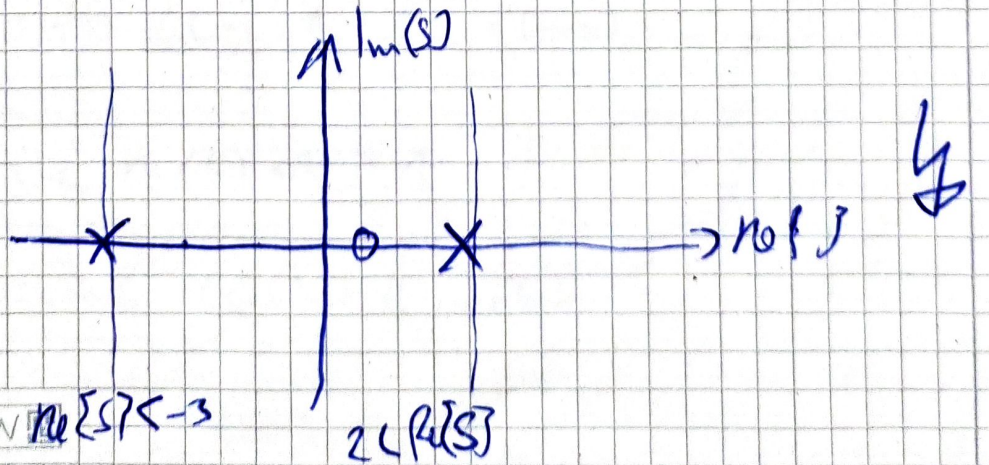
$= \frac{2\left(s+\frac{5}{2}\right)}{(s+2)(s+3)}$



b) $X_a(s) = \frac{1}{s+3} + \frac{1}{s-2} = \frac{2s+1}{(s+3)(s-2)} = \frac{2\left(s+\frac{1}{2}\right)}{(s+3)(s-2)}$



c) $X_a(s) = \frac{1}{s-2} + \frac{1}{s+3} = \frac{2\left(s+\frac{1}{2}\right)}{(s+3)(s-2)}$



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Sig Signale & Systeme

Laplace & Inverse Laplace Transf.

$$X(s) = \frac{2s+4}{s^2+4s+3} = \frac{2s+4}{(s+1)(s+3)}$$

$$\frac{2s+4}{(s+1)(s+3)} = \frac{A}{s+1} + \frac{B}{s+3} \Leftrightarrow 2s+4 = A(s+3) + B(s+1)$$

$s=-1$:

$$2 = A \cdot 2 \rightsquigarrow A = 1$$

$s=-3$:

$$-2 = B(-2) \rightsquigarrow B = 1$$

$$X(s) = \frac{1}{s+1} + \frac{1}{s+3}$$

Inverse Laplace!

d) für $Re(s) > -1$

$$x(t) = e^{-t}\varepsilon(t) + e^{-3t}\varepsilon(t)$$

e) für $Re(s) < -3$

$$x(t) = -e^{-t}\varepsilon(-t) - e^{-3t}\varepsilon(-t)$$

f) $-3 < Re(s) < -1$

$$x(t) = -e^{-t}\varepsilon(-t) + e^{-3t}\varepsilon(t)$$

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Signale & Systeme

Stability of LTI systems

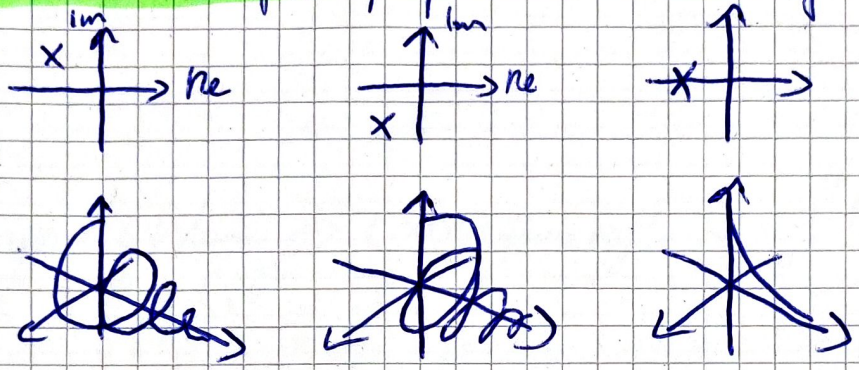
$|x(t)| < \infty \quad \Rightarrow \quad |y(t)| < \infty$ BIBO

$\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$

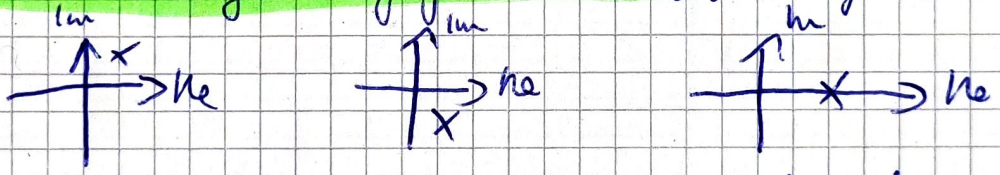
Causal Exponential Functions

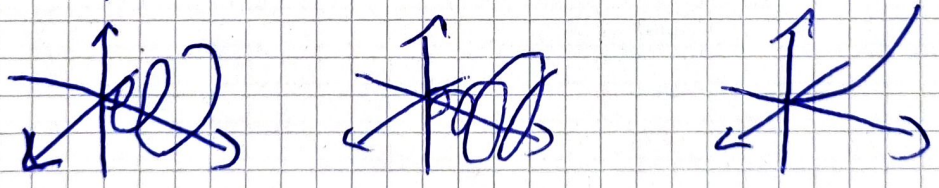
$h(t) = e^{at}\varepsilon(t) = e^{\mathrm{Re}\{a\}t} e^{j\mathrm{Im}\{a\}t} \varepsilon(t) \circ\!\!-\!\!\bullet\, H(s) = \frac{1}{s-a}$

Poles in the left half-plane → Stable system

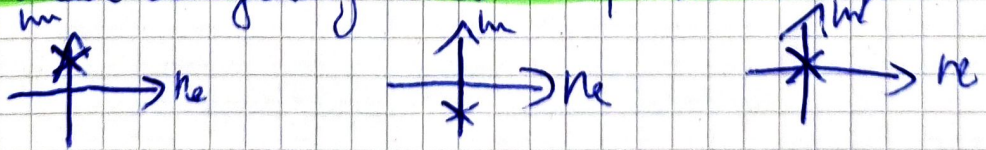


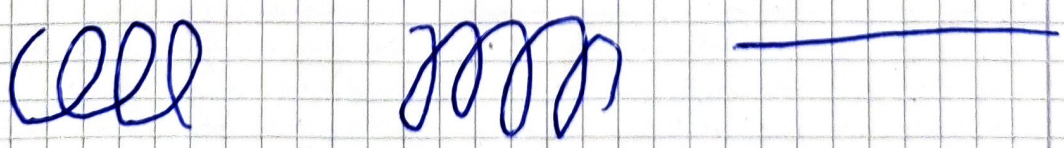
Poles in the right half-plane → unstable system





Poles on imaginary axis ⇒ quasistable system

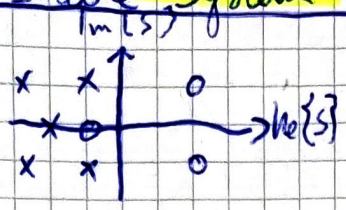




Signale & Systeme

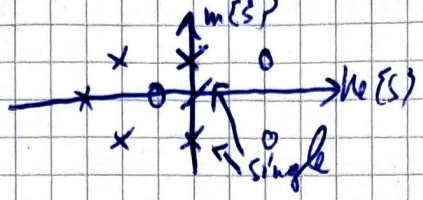
Stabilität of LTI-Systems

Stable System



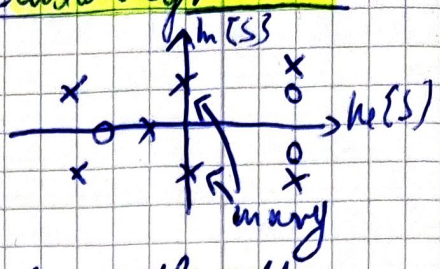
All poles in the left half plane

Quasistable System



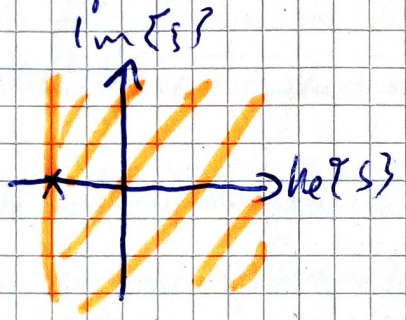
All poles in the left half plane or single poles on the imaginary axis

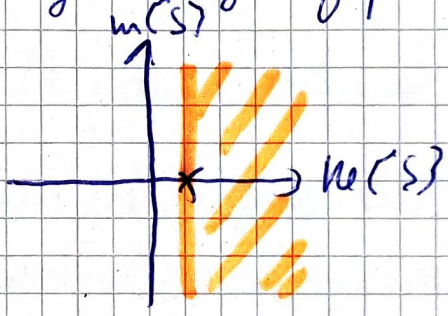
Unstable System



Poles in the right halfplane or m-ary poles on the imginary axis

Causal system → convergence region on right half plane





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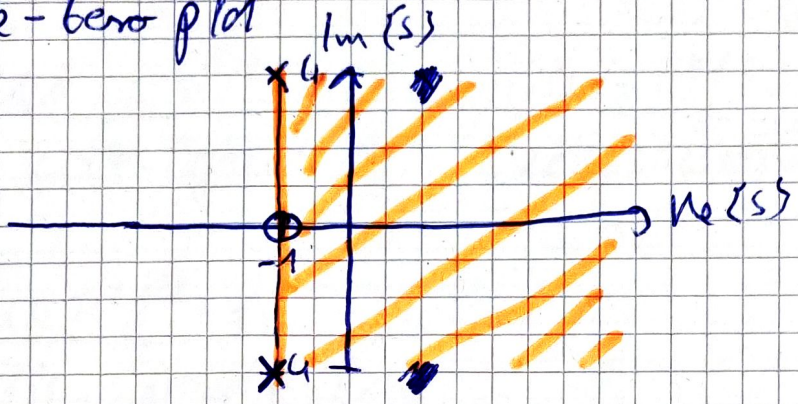
Signale & Systeme

Analog Filters

$$H(s) = \frac{2(s+1)}{s^2+2s+17} = \frac{2(s+1)}{(s-(-1+j4))(s-(-1-j4))}$$

$$s_{1,2} = -1 \pm \sqrt{1^2-17} = -1 \pm j4$$

a) Pole-Zero plot



b) sketch the convergence area of the causal system

A causal system has a causal impulse respone. Such a causal signal is allways associated with a convergence region of a right half-plane

c) Stable → all poles are in the left half-plane

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Signale & Systeme

Analog Filter

d) Calculate $|H(f)|^2$

$$|H(f)|^2 = |H(s)|_{j2\pi f}|^2 = \left| \frac{2(j2\pi f + 1)}{(j2\pi f)^2 + j4\pi f + 17} \right|^2$$

$\hookrightarrow -4\pi^2 f^2$

$$= \frac{16\pi^2 f^2 + 4}{17 - 4\pi^2 f^2 + 16\pi^2 f^2} = \frac{4(4\pi^2 f^2 + 1)}{289 - 34(2\pi f)^2 + (2\pi f)^4 + 4(2\pi f)^2}$$

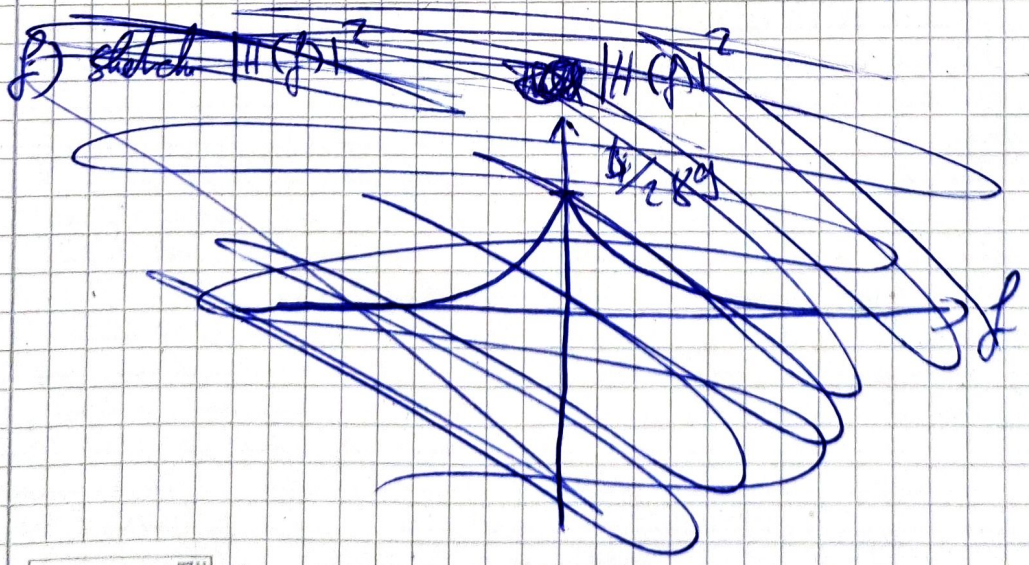
$$= \frac{4(4\pi^2 f^2 + 1)}{(2\pi f)^4 - 30(2\pi f)^2 + 289}$$

e) Calculate $|H(f)|^2$ for $f = 0$ & $f \to \infty$

$$|H(f=0)|^2 = \frac{4}{289}$$

$$|H(f\to\infty)|^2 = \lim_{f\to\infty} \frac{f^4}{f^4} \frac{4(4\pi^2 f^2 + \frac{1}{f^4})}{(2\pi)^4 - 30\frac{(2\pi)^2}{f^2} + \frac{289}{f^4}} \quad f\to\infty = 0$$

f) sketch $|H(f)|^2$



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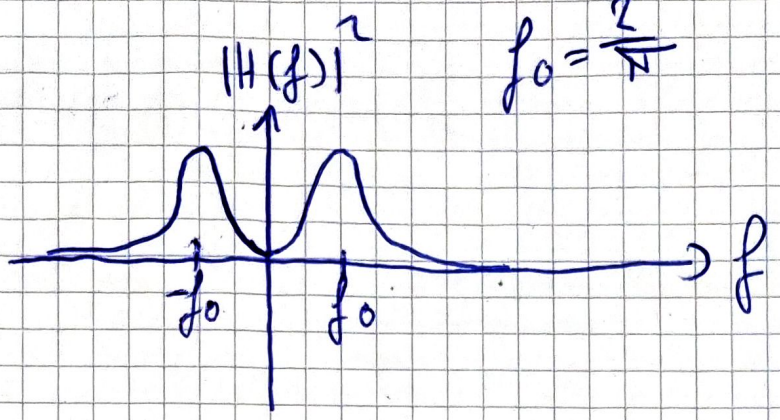
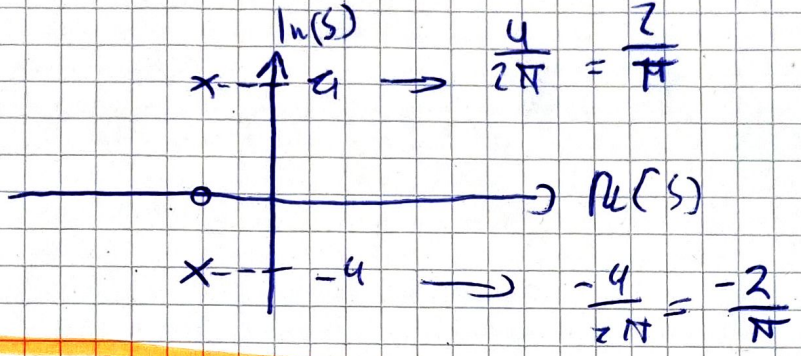
Signale & Systeme

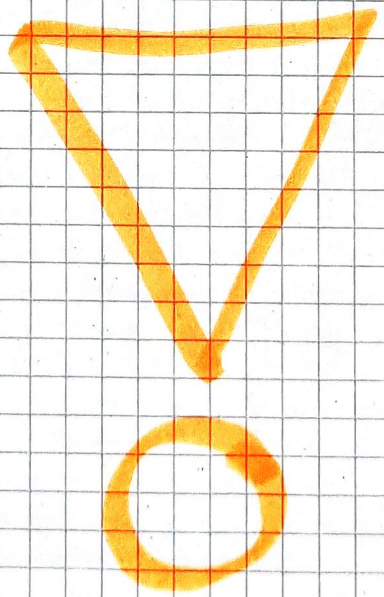
Analog Filters

e)





$$\frac{\ln(s_{\infty,1/2})}{2\pi} = \pm \frac{2}{\pi}$$



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Analog Filters

$$H(s) = \frac{2(s+1)}{(s-(-1+j4))(s-(-1-j4))}$$

$$\frac{2(s+1)}{(s-(-1+j4))(s-(-1-j4))} = \frac{A}{(s-(-1+j4))} + \frac{B}{(s-(-1-j4))}$$

$$2(s+1) = A(s-(-1-j4)) + B(s-(-1+j4))$$

$s = -1+j4$

$$2(-1+j4+1) = A(-1+j4+1+j4)$$

$$j8 = A\,j8 \rightarrow A = 1$$

$s = -1-j4$

$$-j8 = -Bj8 \rightarrow B = 1$$

g) $$H(s) = \frac{1}{(s-(+j4-1))} + \frac{1}{(s-(-1-j4))}$$

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Signale & Systeme

Analog Filter

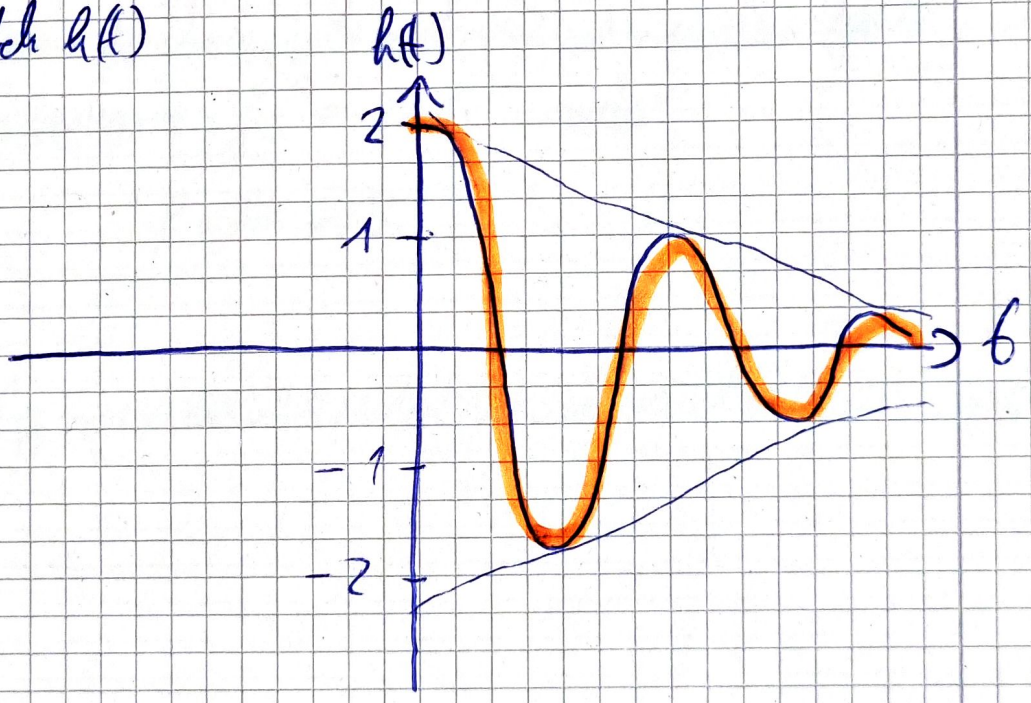
g) ~~$h(t) = e^{(-1+j4)t}\varepsilon(t) + e^{(-1-j4)t}\varepsilon(t)$~~

~~h)~~

$$h(t) = e^{(-1+j4)t}\varepsilon(t) + e^{(-1-j4)t}\varepsilon(t) = e^{-t}\varepsilon(t)\left(e^{+j4t} + e^{-j4t}\right)$$

$$= 2e^{-t}\varepsilon(t)\cos(4t)$$

i) Sketch $h(t)$



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